Rapido Ride Analysis Dashboard

Project Overview

This project analyzes Rapido ride booking data to understand ride performance, revenue trends, customer behavior, cancellation patterns, and payment preferences. The dashboard was developed using Power BI, while SQL was used for data cleaning and transformation.

Tools Used

- SQL
- Power BI
- Microsoft Excel
- Kaggle Dataset

Dataset Information

The dataset contains 50,000 ride records and 13 columns.

Columns Used

- Services
- Date
- Time
- Ride Status
- Source
- Destination
- Duration
- Ride ID
- Distance
- Ride Charge
- Misc Charge
- Total Fare
- Payment Method

Data Cleaning Process

Before creating the dashboard, the dataset was cleaned to improve data quality and reporting accuracy.

1. Missing Value Detection

Checked all columns for null values.

2. Handling Missing Fare Information

The following columns contained missing values:

- Ride Charge
- Misc Charge
- Total Fare
- Payment Method

Approximately 5,036 records contained missing values in these fields.

3. Data Type Validation

Verified and corrected data types:

- Date converted to Date format
- Fare columns converted to Numeric format
- Distance converted to Numeric format
- Duration standardized for analysis

4. Duplicate Check

Ride IDs were checked to ensure each ride record was unique.

5. Data Consistency Validation

Verified:

- Total Fare values
- Service categories
- Ride Status categories
- Payment Method categories

6. Standardization

Cleaned text values to maintain consistency across:

- Service Types
- Payment Methods
- Ride Statuses

SQL Queries Used

1. Retrieve All Successful Rides

SELECT *

FROM rapido

WHERE ride_status = 'Completed';

Purpose

This query extracts all successfully completed rides from the dataset.

Business Value

Completed rides represent actual revenue-generating transactions and are useful for performance analysis.

2. Average Ride Distance by Service Type

SELECT services,

    AVG(distance) AS avg_distance

FROM rapido

GROUP BY services;

Purpose

Calculates the average distance traveled for each service category.

Business Value

Helps identify customer travel patterns across Bike, Auto, and Cab services.

3. Total Cancelled Rides

SELECT COUNT(ride_status)

FROM rapido

WHERE ride_status = 'Cancelled';

Purpose

Counts the total number of cancelled rides.

Business Value

Cancellation rates help measure operational efficiency and customer satisfaction.

4. Top 5 Ride IDs with Highest Completed Rides

```
SELECT ride_id,
    COUNT(ride_id) AS completed_rides
FROM rapido
WHERE ride_status = 'Completed'
GROUP BY ride_id
ORDER BY completed_rides DESC
LIMIT 5;
```

Purpose

Identifies ride records with the highest completion count.

Business Value

Useful for understanding ride activity and validating ride transaction data.

5. Total Revenue Generated

```
SELECT SUM(total_fare) AS total_revenue
FROM rapido;
```

Purpose

Calculates total revenue generated from all rides.

Business Value

Provides an overall measure of business performance.

6. Revenue by Service Type

```
SELECT services,
    SUM(total_fare) AS total_revenue
FROM rapido
GROUP BY services
ORDER BY total_revenue DESC;
```

Purpose

Calculates revenue contribution by each service category.

Business Value

Identifies the most profitable service segment.

7. Retrieve All Paytm Transactions

```
SELECT *

FROM rapido

WHERE payment_method = 'Paytm';
```

Purpose

Extracts rides where payment was made using Paytm.

Business Value

Helps analyze customer payment preferences and digital payment adoption.

8. Top 10 Pickup Locations

```
SELECT source,

    COUNT(source) AS total_rides

FROM rapido

GROUP BY source

ORDER BY total_rides DESC

LIMIT 10;
```

Purpose

Identifies locations with the highest ride demand.

Business Value

Supports demand forecasting and service allocation decisions.

9. Top 10 Longest Rides

```
SELECT *

FROM rapido
```

```
ORDER BY duration DESC

LIMIT 10;
```

Purpose

Retrieves rides with the longest travel duration.

Business Value

Helps understand long-distance travel behavior and fare contribution.

10. Average Trip Distance

```
SELECT AVG(distance) AS average_distance

FROM rapido;
```

Purpose

Calculates the average trip distance across all rides.

Business Value

Provides an overview of customer travel patterns and service utilization.

Dashboard KPIs

The dashboard includes the following Key Performance Indicators (KPIs):

- Total Rides
- Total Revenue
- Average Fare
- Average Distance
- Cancellation Rate
- Revenue by Service Type
- Revenue Trend Over Time
- Payment Method Distribution
- Ride Status Distribution

Key Insights

1. Ride bookings show clear trends across different service categories.
2. Revenue generation varies significantly between service types.
3. Digital payment methods dominate ride transactions.
4. Cancellation rates impact overall operational efficiency.
5. Distance and duration strongly influence fare generation.

6. Revenue trends help identify peak demand periods.

Conclusion

This project demonstrates the complete data analytics workflow, including data cleaning, SQL querying, KPI creation, and dashboard development in Power BI. The analysis provides actionable insights into ride performance, revenue generation, customer preferences, and operational efficiency.

Dashboard Screenshot